DR. MARIA BEATRIZ WALTER COSTA

EU permanent resident (Swiss-Brazilian)
waltercostamb@gmail.com
https://waltercostamb.github.io/

SUMMARY

Bioinformatician with **PhD** and **15+ years**' experience in **analysis of large-scale biological datasets**. Proven track record in generating biologically meaningful insights to **support data-driven decisions**. Published author (*BMC Bioinformatics, JMIR Medical Informatics*) with advanced expertise in **Python, R and statistical modeling**.

WORK EXPERIENCE

Research Associate (fixed-term) — Jena, Germany
Friedrich-Schiller University Jena — *10/2022 - 09/2025*

- Led a project on large-scale microbial omics data to identify environmental biomarkers (ML, Python)
- Developed a scalable, reproducible high-throughput pipeline for genomics data (Snakemake, Slurm)
- Identified bacterial biomarkers associated with temperature, with potential biotechnological applications
- Co-supervised a doctoral researcher and contributed to teaching and training in large-scale data analysis

Research Associate (fixed-term) — Leipzig, Germany
University of Leipzig Medical Center — *02/2020 - 12/2021*

- Delivered data-driven approaches to support automated diagnosis of acute conditions (Statistics, R)
- Worked closely with physicians and industry partners to build AMPEL Clinical Decision Support System
- Generated insights from complex data to support clinical decision-making and improve patient safety
- Structured a clinical knowledge dataflow to improve downstream analysis (SAP-based data)

Consultant (fixed-term) — Brasília/DF, Brazil
Brazilian Agricultural Research Corporation (EMBRAPA)/Agroenergy — *11/2018 - 03/2019*

- Analyzed differential expression data to find key genes that optimize ethanol production (R, DESeq2)
- Collaborated with experimental scientists to support decision-making in strain engineering

Research Associate (fixed-term) — Leipzig, Germany
University of Leipzig — *04/2017 - 03/2018*

- Led an omics-focused project to improve understanding of ncRNA evolution (Bioinformatics, Perl)
- Identified ncRNA biomarkers associated with cognitive functions, with potential relevance to diseases

EDUCATION

Dr. rer. nat. in Computer Science (*magna cum laude*) — 2013 - 2018

- University of Leipzig, Faculty of Mathematics and Computer Science, Chair of Bioinformatics (Germany)
- Specialization in method development in Computational Biology

M. Sc. in Molecular Biology 2010 - 2012

- University of Brasília, Institute of Biological Sciences, Graduation Program of Molecular Biology (Brazil)
- Specialization in NGS-based analysis in Immunology

B. Sc. in Biological Sciences 2006 - 2010

- University of Brasília, Institute of Biological Sciences (Brazil)

LANGUAGES

English: near-native speaker (C2, Cambridge CPE)
Portuguese: Native speaker (C2)
German: Advanced (C1)
Spanish: Advanced (B2/C1)

TECHNICAL SKILLS

Programming: Python, R, Perl, Bash, SQL (mySQL), AI-assisted coding (Gemini)
Bioinformatics: genomics, transcriptomics, AI/ML: machine learning, statistical modeling
Workflow: Snakemake, Git (GitHub)
Data Processing: Linux/Unix, HPC platforms (Slurm, SGE)

SELECTED PRODUCTS

- `FxTractor`: reproducible high-throughput pipeline for feature extraction from microbial genomes (150k+ features across 13k+ genomes), leveraging parallel computing on HPC platforms

  `https://github.com/MGXlab/FxTractor`

- `SSS-test`: the first tool of the scientific community that detects adaptive selection in ncRNAs, demonstrated by a publication in *BMC Bioinformatics*

  `https://github.com/waltercostamb/SSS-test`

SELECTED PUBLICATIONS

- **Walter Costa MB** *et al.* The Clinical Decision Support System AMPEL for Laboratory Diagnostics: Implementation and Technical Evaluation. *JMIR Med Informatics*, 2021

  `https://medinform.jmir.org/2021/6/e20407`

- **Walter Costa, MB** *et al.* SSS-Test: A novel test for detecting selection on the secondary structures of non-coding RNAs. *BMC Bioinformatics*, 2019

  `https://doi.org/10.1186/s12859-019-2711-y`